

Non-conformal obstructions to bubble expansion

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Introduction

First-order phase transitions (FOPTs) offer a unique probe of physics beyond the Standard Model, occurring through the nucleation, expansion, and collision of bubbles—a process that could generate gravitational waves in the early universe or in neutron star mergers. A key parameter in modeling FOPTs is the speed of sound: while often assumed constant [1], transitions involving many degrees of freedom show a strong temperature dependence. We study self-similar bubble solutions for a broad class of non-conformal holography-inspired equations of state (EOS), identifying novel hydrodynamic obstructions that produce gaps between deflagration, hybrid, and detonation regimes.

Hydrodynamics

Late time dynamics of a single bubble expansion can be described by the conservation of energy momentum tensor of an ideal fluid

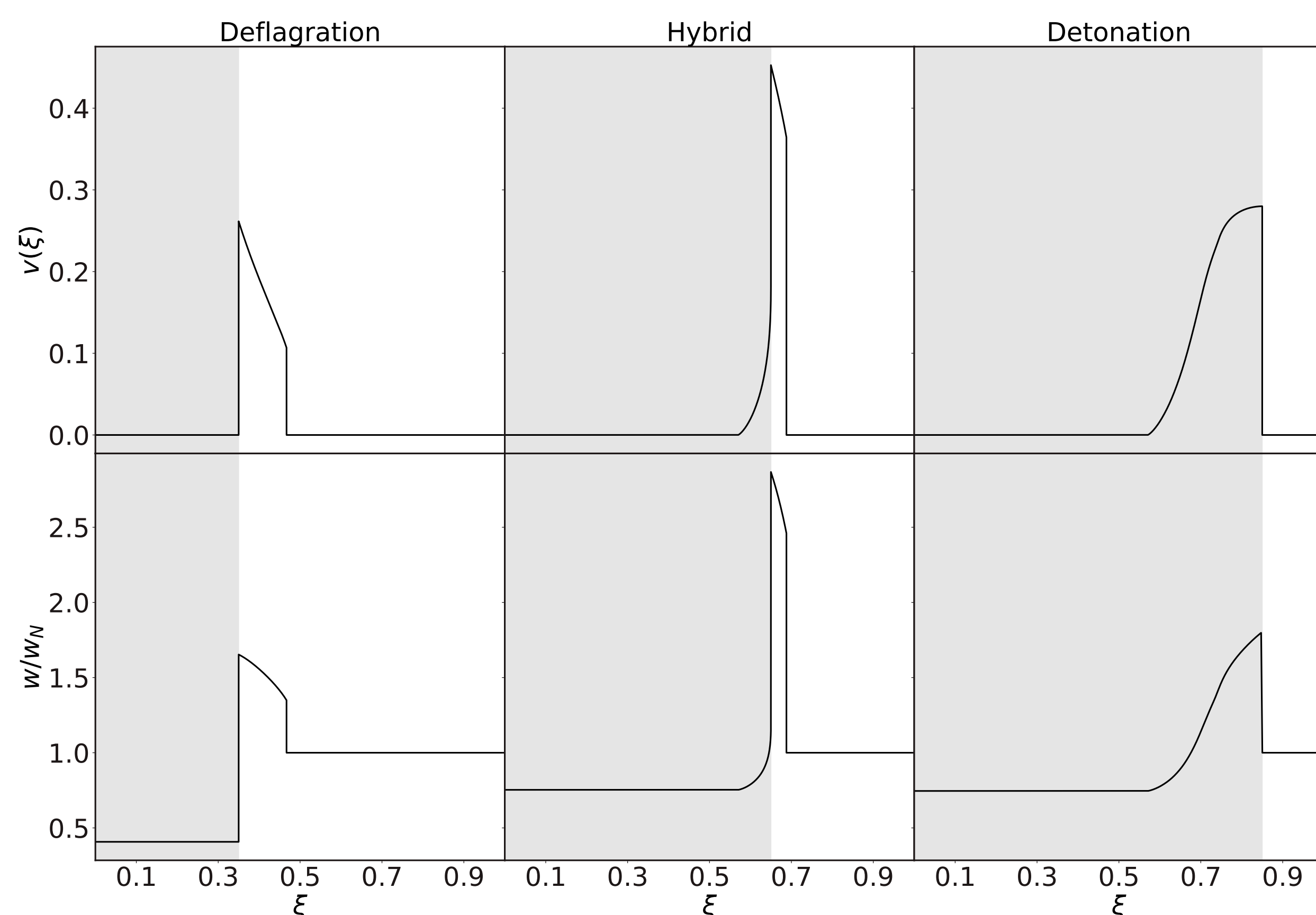
$$\nabla_\mu T^{\mu\nu} = 0, \quad T^{\mu\nu} = w u^\mu u^\nu + p \eta^{\mu\nu}$$

where $w = e + p$ and $u^\mu = \gamma(1, \vec{v})$, where γ is the Lorentz factor. The only microscopic information needed is the EOS, $p(e)$, with speed of sound $c_s^2(e) = \partial_e p$. We seek for self-similar solutions $\xi = r/t$ by solving for $v(\xi)$ and $e(\xi)$ with possible discontinuities

$$\left(\begin{aligned} (\xi - v) \frac{\partial_\xi e}{e + p} &= 2 \frac{v}{\xi} + (1 - \gamma^2 v(\xi - v)) \partial_\xi v, \\ c_s^2(1 - v\xi) \frac{\partial_\xi e}{e + p} &= \gamma^2(\xi - v) \partial_\xi v \end{aligned} \right) \left| \begin{aligned} v_+ v_- &= \frac{p_+ - p_-}{e_+ - e_-}, \\ \frac{v_+}{v_-} &= \frac{e_- + p_+}{e_+ + p_-} \end{aligned} \right.$$

where $\pm$ denotes ahead and behind the discontinuity. We find three type of solutions:

- Deflagrations: bubble wall velocity smaller than the speed of sound
- Detonations: bubble wall velocity larger than the speed of sound
- Hybrids: fluid moves just at the speed of sound at the wall.

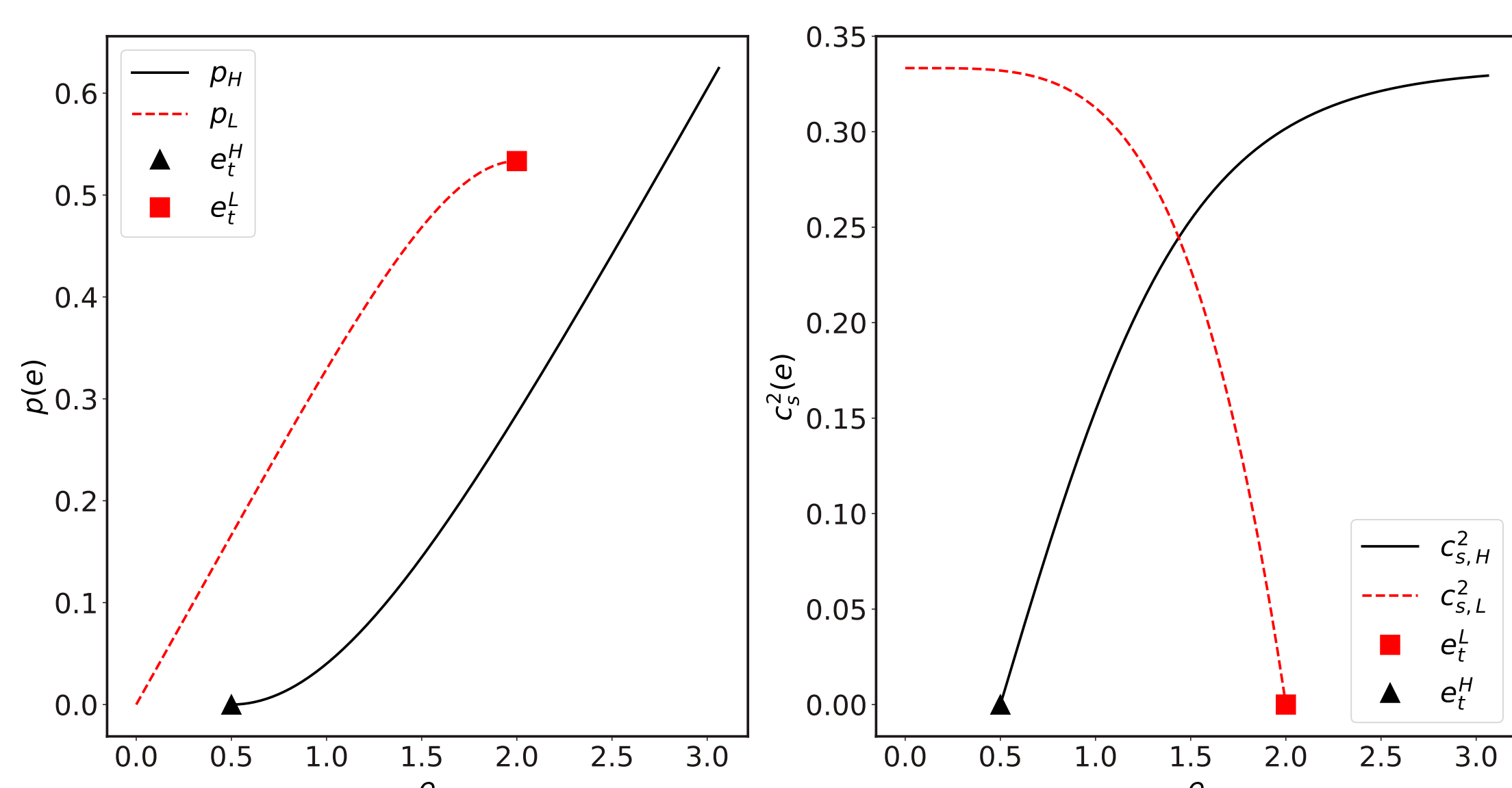


Equation of state

We choose a parametrization for the EOS fulfilling

$$\left. \frac{dp_L}{de} \right|_{e=e_t^L} = \left. \frac{dp_H}{de} \right|_{e=e_t^H} = 0, \quad \lim_{e \rightarrow 0} \frac{dp_L}{de} = \lim_{e \rightarrow \infty} \frac{dp_H}{de} = \frac{1}{3},$$

$$p_L(e) = \frac{1}{3} \left(e - \frac{e^{n+1}}{(n+1)(e_t^L)^n} \right), \quad p_H(e) = \frac{1}{3} \delta \log \left[\cosh \left(\frac{e - e_t^H}{\delta} \right) \right] + p_1.$$

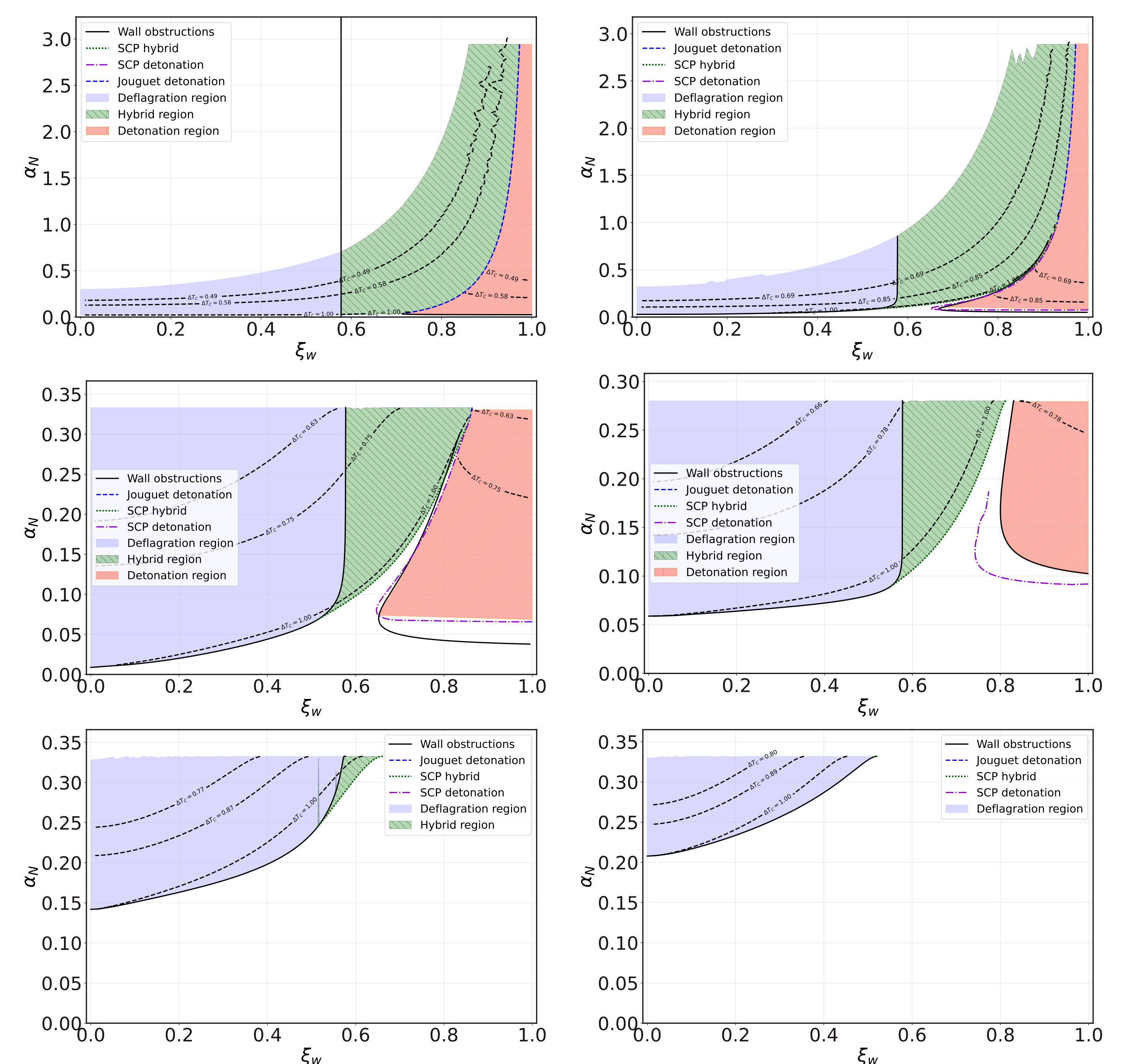


Results

We consider different EOS interpolating from a bag model like (Electroweak-type theories) to a holographically motivated one. We construct all solutions as a function of the bubble wall velocity ξ_w and the transition strength $\alpha(T)$

$$\alpha(T) = \frac{4}{3} \frac{\theta_H - \theta_L}{e_H + p_H} \Big|_T, \quad \theta = \frac{e + 3p}{4}$$

where H and L denote high and low energy phases respectively.



We found two type of obstructions to bubble expansion

- Wall obstructions (black line): deflagrations are limited by the subsonic wall condition. Detonations are limited because there is a finite amount of superheating.

$$\text{Deflagrations: } \xi_w \leq c_s(e_-), \quad \text{Detonations: } e \leq e_t^L$$

- Flow obstructions: restrictions involving the rarefaction wave. They affect both detonations and hybrids. The flow becomes multivalued if the energy is larger than the critical sound point (SCP):

$$\left(\frac{\xi - v}{1 - v\xi} \right)^2 = c_s^2(e), \quad \frac{dc_s^2}{de} = - \frac{2(1 - c_s^2(e))c_s^2(e)}{e + p(e)}$$

When going from a bag model-like EOS to a holographically motivated one, a gap appears between hybrids and detonations. This gap increases until detonations disappear. If one pushes more the EOS, hybrids will disappear too.

References

- [1] Energy Budget of Cosmological First-order Phase Transitions
Jose R. Espinosa, Thomas Konstandin, Jose M. No, Geraldine Servant
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